IGCSE 0607 Practice Questions — E2.2 — Combined Questions & Markscheme

Algebraic manipulation

Total Questions: 30 — **Total Marks:** 168 Marks

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- **E2.2** Algebraic manipulation
  → **Paper 2 (Non-Calculator)** *(24 Qs)* · **Paper 4 (Calculator)** *(6 Qs)*

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1 Expand and simplify.

$(3x - 2)(2x + 5)$

..... **[2]**

2 **(a)** Expand and simplify.

$3(2x - 1) - 2(x + 4)$

..... **[2]**

(b) Factorise fully.

$15x^2y - 20xy^2$

..... **[2]**

3 **(a)** Expand and simplify.

$4(2a - 3b) - 3(a - 5b)$

..... **[2]**

(b) Factorise completely.

$12x^2y - 18xy^2$

..... **[2]**

4 **(a)** Expand and simplify.

$4(2x - 3y) - 3(x - 5y)$

..... **[2]**

(b) Factorise completely.

$14a^2b - 21ab^2$

..... **[2]**

5 **(a)** Expand and simplify

$$4(3x - 2) - 2(x - 5)$$

..... **[2]**

(b) Factorise fully

$$15a^2b - 10ab^2$$

..... **[2]**

6 **(a)** Expand and simplify.

$3(2x - 5) - 2(x + 4)$

..... **[2]**

(b) Factorise completely.

$8x^2y - 12xy^2$

..... **[2]**

7 **(a)** Simplify

$$4x^2 - 7xy + 3 - 9x^2 + 2xy - 8$$

..... **[2]**

(b) Factorise fully

$$14a^3b - 21a^2b^2$$

..... **[2]**

8 **(a)** Expand and simplify

$$3(2x - 5) - 4(x - 2)$$

..... **[2]**

(b) Factorise fully

$$12x^2y - 18xy^2$$

..... **[2]**

9 **(a)** Simplify.

$3a - 5b + 2a + 8b$

..... **[2]**

(b) Factorise completely.

$12x^2y - 18xy^2$

..... **[2]**

10 **(a)** Expand and simplify.

$3(2x - 5) - 4(x - 3)$

..... **[2]**

(b) Factorise fully.

$14x^2y + 21xy^2$

..... **[2]**

11 **(a)** Factorise completely.

$12x^2 - 27y^2$

..... **[2]**

(b) Factorise completely.

$2ax - 6ay + bx - 3by$

..... **[2]**

12 **(a)** Expand and simplify.

$(2x - 3)(x + 5)$

..... **[2]**

(b) Factorise.

$y^2 - 16x^2$

..... **[1]**

(c) Factorise.

$p^2 - 4pq - 5q^2$

..... **[2]**

13 **(a)** Expand and simplify.

$(2x - 3)(x + 5)$

..... **[2]**

(b) Factorise.

$ac - 3bc + 2ad - 6bd$

..... **[2]**

(c) Factorise.

$16x^2 - 25y^2$

..... **[1]**

14 **(a)** Expand and simplify.

$(2x - 3)(x + 4)$

..... **[2]**

(b) Factorise completely.

$ax - bx - 3ay + 3by$

..... **[2]**

(c) Factorise.

$49x^2 - 16y^2$

..... **[2]**

15 **(a)** Factorise.

$x^2 - 8x + 15$

..... **[2]**

(b) Factorise.

$2x^2 - 5x - 12$

..... **[2]**

(c) Factorise.

$ax - 3a + 2bx - 6b$

..... **[2]**

16 **(a)** Expand and simplify

$$(3x-2)(x+5)-(2x-3)^2$$

..... **[3]**

(b) Factorise completely:

(i) $3x^3-12xy^2$

..... **[2]**

(ii) $6x^2-11x-10$

..... **[2]**

17 **(a)** Expand and simplify.

$(2x - 5)(3x + 1)$

..... **[2]**

(b) Factorise.

$x^2 - 7x - 18$

..... **[2]**

(c) Factorise completely.

$18x^2 - 50y^2$

..... **[3]**

18 **(a)** Expand and simplify.

$(x-2)(2x+3)(x-5)$

..... **[3]**

(b) Factorise completely.

$12x^3 + 17x^2 - 5x$

..... **[3]**

19 **(a)** Factorise completely.

$18x^2 - 50y^2$

..... **[3]**

(b) Factorise completely.

$3x^3 + 10x^2 - 8x$

..... **[3]**

20 **(a)** Expand and simplify.

$$(x-2)(x+3)(2x-1)$$

..... **[3]**

(b) Factorise completely.

(i)

$$6x^2 - 11x - 10$$

..... **[2]**

(ii)

$$ax - ay + 3cx - 3cy$$

..... **[2]**

21 **(a)** Factorise

$$cx - 5dx + 3cy - 15dy$$

..... **[2]**

(b) Factorise fully

$$2x^3 - 18x$$

..... **[2]**

(c) Factorise fully

$$12x^3 - 14x^2 - 10x$$

..... **[3]**

22 **(a)** Expand and simplify.

$(x-4)(2x+1)(x+3)$

..... **[3]**

(b) Factorise completely.

$2ax - 3by + 2ay - 3bx$

..... **[2]**

(c) Factorise.

$6x^2 + 7x - 20$

..... **[2]**

23 **(a)** Expand and simplify

$$(2x-1)(x+3)(3x-2)$$

..... **[3]**

(b) Factorise completely

$$4ax-6ay-6bx+9by$$

..... **[2]**

(c) Given that $x^2-y^2=45$ and $x-y=5$, find the value of $(x+y)^2$.

..... **[3]**

24 **(a)** Factorise

$$6ax - 2ay - 3bx + by$$

..... **[2]**

(b) Factorise completely.

(i)

$$50x^2 - 8y^2$$

..... **[3]**

(ii)

$$2x^3 + 5x^2 - 12x$$

..... **[3]**

25 **(a)** Expand and simplify

$$(x-3)(2x+1)(x+4)$$

..... **[3]**

(b) Factorise

$$2x^2-7x-15$$

..... **[2]**

26 **(a)** Expand and simplify.

$(2x - 3)(x + 5)$

..... **[2]**

(b) Factorise.

$x^2 - 7x - 18$

..... **[2]**

(c) Factorise completely.

$4x^2 - 25y^2$

..... **[2]**

27 **(a)** Expand and simplify

$$(x-4)(2x+3)(x+2)$$

..... **[3]**

(b) Factorise

$$16x^2 - 49y^2$$

..... **[2]**

(c) Factorise

$$3x^2 - 10x - 8$$

..... **[2]**

28 **(a)** Expand and simplify.

$(x-4)(2x+3)(x+1)$

..... **[3]**

(b) Factorise completely.

$6x^2 - 11x - 10$

..... **[2]**

(c) Factorise completely.

$x^3 + 3x^2 - 4x - 12$

..... **[3]**

29 **(a)** Expand and simplify.

$(x-2)(2x+1)(x+3)$

..... **[3]**

(b) Factorise.

$6x^2 - 11x - 10$

..... **[2]**

(c) Factorise completely.

$3x^3 + 5x^2 - 2x$

..... **[3]**

30 **(a)** Expand and simplify.

$(x-2)(x+3)(2x-1)$

..... **[3]**

(b) Factorise.

$6x^2 - 11x - 10$

..... **[2]**

(c) Factorise completely.

$2x^3 - 18x$

..... **[3]**

1 At least 3 terms correct from $6x^2 + 15x - 4x - 10$ **[M1]**

$6x^2 + 11x - 10$ **[A1]**

[Back to Question] *[Total Marks: 2]*

2 (a) Expansion to $6x - 3 - 2x - 8$ **[M1]**

$4x - 11$ **[A1]**

(b) Extracting a partial common factor, e.g. $5xy(3x - 4y)$ or $x(15xy - 20y^2)$ or $5(3x^2y - 4xy^2)$ **[M1]**

$5xy(3x - 4y)$ **[A1]**

[Back to Question] *[Total Marks: 4]*

3 (a) M1 for $8a - 12b$ or $-3a + 15b$ seen

$5a + 3b$ **[A1]**

(b) M1 for a correct partial factorisation, e.g., $6x(2xy - 3y^2)$ or $2xy(6x - 9y)$

$6xy(2x - 3y)$ **[A1]**

[Back to Question] *[Total Marks: 4]*

4 (a) $8x - 12y - 3x + 15y$ **[M1]**

$5x + 3y$ **[A1]**

(b) $7ab$ extracted as a common factor **[M1]**

$7ab(2a - 3b)$ **[A1]**

[Back to Question] *[Total Marks: 4]*

5 (a) $12x - 8 - 2x + 10$ **[M1]**

$10x + 2$ **[A1]**

(b) $5ab(3a - 2b)$ **[B2]**

If not B2, then B1 for $5a(3ab - 2b^2)$ or $5b(3a^2 - 2ab)$ or $ab(15a - 10b)$ **[B1]**

[Back to Question] *[Total Marks: 4]*

6 (a) $6x - 15 - 2x - 8$ **[M1]**

$4x - 23$ **[A1]**

(b) $4xy(2x - 3y)$ or $4x(2xy - 3y^2)$ or $4y(2x^2 - 3xy)$ or $xy(8x - 12y)$ **[M1]**

$4xy(2x - 3y)$ **[A1]**

[Back to Question] *[Total Marks: 4]*

7 **(a)** Collects at least two like terms correctly: $-5x^2$, $-5xy$, or -5 **[M1]**

$-5x^2 - 5xy - 5$ **[A1]**

(b) Extracts partial common factor correctly, e.g. $7a(2a^2b - 3ab^2)$ or $a^2b(14a - 21b)$ **[M1]**

$7a^2b(2a - 3b)$ **[A1]**

[Back to Question] *[Total Marks: 4]*

8 **(a)** Correct expansion: $6x - 15 - 4x + 8$ **[M1]**

$2x - 7$ **[A1]**

(b) $6xy(2x - 3y)$ **[B2]**

Correct partial factorisation e.g., $6x(2xy - 3y^2)$ or $2xy(6x - 9y)$ **[M1]**

[Back to Question] *[Total Marks: 4]*

9 **(a)** $5a$ or $+3b$ **[B1]**

$5a + 3b$ **[A1]**

(b) $6xy(2x - 3y)$ **[B2]**

If B0, allow [M1] for a correct partial factorisation, e.g. $6x(2xy - 3y^2)$ or $3xy(4x - 6y)$

[Back to Question] *[Total Marks: 4]*

10 **(a)** Expansion to give $6x - 15$ and $-4x + 12$ **[M1]**

$2x - 3$ **[A1]**

(b) $7xy(2x + 3y)$ **[B2]**

Partial factorisation, e.g. $7x(2xy + 3y^2)$ or $xy(14x + 21y)$ **[B1]**

[Back to Question] *[Total Marks: 4]*

11 **(a)** $3(4x^2 - 9y^2)$ or $(6x - 9y)(2x + 3y)$ or $(2x - 3y)(6x + 9y)$ **[M1]**

$3(2x - 3y)(2x + 3y)$ **[A1]**

(b) $2a(x - 3y) + b(x - 3y)$ or $x(2a + b) - 3y(2a + b)$ **[M1]**

$(2a + b)(x - 3y)$ **[A1]**

[Back to Question] *[Total Marks: 4]*

12 **(a)** $2x^2 + 10x - 3x - 15$ **[M1]**

$2x^2 + 7x - 15$ **[A1]**

(b) $(y - 4x)(y + 4x)$ **[B1]**

(c) $(p + q)(p - 5q)$ **[B2]**

If B0, allow [M1] for $(p+aq)(p+bq)$ where $ab=-5$ or $a+b=-4$

[Back to Question] *[Total Marks: 5]*

13 **(a)** $2x^2+10x-3x-15$ **[M1]**

$2x^2+7x-15$ **[A1]**

(b) $c(a-3b)+2d(a-3b)$ or $a(c+2d)-3b(c+2d)$ **[M1]**

$(a-3b)(c+2d)$ **[A1]**

(c) $(4x-5y)(4x+5y)$ **[B1]**

[Back to Question] *[Total Marks: 5]*

14 **(a)** M1 for 3 correct terms out of $2x^2+8x-3x-12$

$2x^2+5x-12$ **[A1]**

(b) M1 for $x(a-b)$ or $-3y(a-b)$ seen

$(x-3y)(a-b)$ **[A1]**

(c) $(7x-4y)(7x+4y)$ **[B2]**

If 0 scored, allow M1 for $(7x-4y)(7x-4y)$ or $(7x+4y)(7x+4y)$

[Back to Question] *[Total Marks: 6]*

15 **(a)** $(x+a)(x+b)$ where $ab=15$ or $a+b=-8$ **[M1]**

$(x-3)(x-5)$ **[A1]**

(b) $(2x+a)(x+b)$ where $ab=-12$ or $2b+a=-5$ **[M1]**

$(2x+3)(x-4)$ **[A1]**

(c) $a(x-3)+2b(x-3)$ or $x(a+2b)-3(a+2b)$ **[M1]**

$(a+2b)(x-3)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

16 **(a)** $(3x-2)(x+5)=3x^2+13x-10$ **[B1]**

$(2x-3)^2=4x^2-12x+9$ **[B1]**

$-x^2+25x-19$ **[B1]**

(b) **(i)** $3x(x^2-4y^2)$ **[M1]**

$3x(x-2y)(x+2y)$ **[A1]**

(ii) $(3x+a)(2x+b)$ where $ab=-10$ or $3b+2a=-11$ **[M1]**

$(3x+2)(2x-5)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

17 **(a)** $6x^2 + 2x - 15x - 5$ (at least 3 correct terms) **[M1]**

$6x^2 - 13x - 5$ **[A1]**

(b) $(x + a)(x + b)$ where $ab = -18$ or $a + b = -7$ **[M1]**

$(x - 9)(x + 2)$ **[A1]**

(c) $2(9x^2 - 25y^2)$ **[M1]**

$(3x - 5y)(3x + 5y)$ seen **[M1]**

$2(3x - 5y)(3x + 5y)$ **[A1]**

[Back to Question] *[Total Marks: 7]*

18 **(a)** Correct expansion of any two brackets, e.g. $2x^2 - x - 6$ or $2x^2 - 7x - 15$ or $x^2 - 7x + 10$ **[M1]**

Correctly multiplies their 3-term quadratic by the remaining bracket to get at least 4 correct terms **[M1]**

$2x^3 - 11x^2 - x + 30$ **[A1]**

(b) Extracts common factor x to give $x(12x^2 + 17x - 5)$ **[M1]**

Correctly factorises the quadratic expression to $(4x - 1)(3x + 5)$ **[M1]**

$x(4x - 1)(3x + 5)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

19 **(a)** Taking out common factor to reach $2(9x^2 - 25y^2)$ **[M1]**

Factorising difference of squares to reach $(3x - 5y)(3x + 5y)$ **[M1]**

$2(3x - 5y)(3x + 5y)$ **[A1]**

(b) Taking out common factor x to reach $x(3x^2 + 10x - 8)$ **[M1]**

Factorising quadratic to reach $(3x - 2)(x + 4)$ **[M1]**

$x(3x - 2)(x + 4)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

20 **(a)** Correct expansion of two brackets, e.g. $(x^2 + x - 6)(2x - 1)$ or $(x - 2)(2x^2 + 5x - 3)$ **[M1]**

Multiplication of quadratic by third bracket yielding 6 terms, e.g. $2x^3 - x^2 + 2x^2 - x - 12x + 6$ **[M1]**

$2x^3 + x^2 - 13x + 6$ **[A1]**

(b) **(i)** $(3x + a)(2x + b)$ where $ab = -10$ or $2a + 3b = -11$ **[M1]**

$(3x + 2)(2x - 5)$ **[A1]**

(ii) $a(x - y) + 3c(x - y)$ or $x(a + 3c) - y(a + 3c)$ **[M1]**

$(a + 3c)(x - y)$ **[A1]**

[Back to Question] *[Total Marks: 7]*

21 **(a)** $x(c-5d)+3y(c-5d)$ or $c(x+3y)-5d(x+3y)$ **[M1]**

$(x+3y)(c-5d)$ **[A1]**

(b) $2x(x^2-9)$ **[M1]**

$2x(x-3)(x+3)$ **[A1]**

(c) $2x(6x^2-7x-5)$ **[M1]**

$2x(3x-5)(2x+1)$ **[A2]**

If not A2, then A1 for $(3x-5)(2x+1)$ or $2x(3x+5)(2x-1)$ **[A1]**

[Back to Question] *[Total Marks: 7]*

22 **(a)** Expands any two brackets, e.g. $(2x^2-7x-4)(x+3)$ **[M1]**

Multiplies their quadratic by the third bracket, e.g. $2x^3-7x^2-4x+6x^2-21x-12$ **[M1]**

$2x^3-x^2-25x-12$ **[A1]**

(b) $2a(x+y)-3b(y+x)$ or $x(2a-3b)+y(2a-3b)$ **[M1]**

$(x+y)(2a-3b)$ **[A1]**

(c) $(2x+5)(3x-4)$ **[B2]**

If B0, allow [M1] for $(2x+a)(3x+b)$ where $ab=-20$ or $3a+2b=7$

[Back to Question] *[Total Marks: 7]*

23 **(a)** Correctly expands any two brackets, e.g. $(2x^2+5x-3)(3x-2)$ **[M1]**

Multiplies by the third bracket correctly: $6x^3-4x^2+15x^2-10x-9x+6$ **[M1]**

$6x^3+11x^2-19x+6$ **[A1]**

(b) $2a(2x-3y)-3b(2x-3y)$ or $2x(2a-3b)-3y(2a-3b)$ **[M1]**

$(2a-3b)(2x-3y)$ **[A1]**

(c) Uses difference of two squares: $(x-y)(x+y)=45$ **[M1]**

$5(x+y)=45 \implies x+y=9$ **[M1]**

$(x+y)^2=9^2=81$ **[A1]**

[Back to Question] *[Total Marks: 8]*

24 **(a)** $(2a-b)(3x-y)$ **[B2]**

Correct partial factorisation e.g., $2a(3x-y)-b(3x-y)$ or $3x(2a-b)-y(2a-b)$ **[M1]**

(b) **(i)** Extracts common factor of 2: $2(25x^2-4y^2)$ **[M1]**

Factorises difference of two squares: $(5x-2y)(5x+2y)$ **[M1]**

$2(5x-2y)(5x+2y)$ **[A1]**

(ii) Extracts common factor of x: $x(2x^2+5x-12)$ **[M1]**

Factorises quadratic: $(2x-3)(x+4)$ **[M1]**

$x(2x-3)(x+4)$ **[A1]**

[Back to Question] *[Total Marks: 8]*

25 **(a)** Correct expansion of any two brackets, e.g., $(2x^2-5x-3)$ or $(2x^2+9x+4)$ or (x^2+x-12) **[M1]**

Correct expansion of their quadratic with the third bracket, producing at least 5 terms, e.g., $2x^3+8x^2-5x^2-20x-3x-12$ **[M1]**

$2x^3+3x^2-23x-12$ **[A1]**

(b) $(2x+3)(x-5)$ **[B2]**

Accept $(2x+a)(x+b)$ where $ab=-15$ or $2b+a=-7$ **[M1]**

[Back to Question] *[Total Marks: 5]*

26 **(a)** Four terms with at least three correct from $2x^2+10x-3x-15$ **[M1]**

$2x^2+7x-15$ **[A1]**

(b) $(x+a)(x+b)$ where $ab=-18$ or $a+b=-7$ **[M1]**

$(x-9)(x+2)$ **[A1]**

(c) $(2x+a)(2x-a)$ or $(cx+5y)(cx-5y)$ **[M1]**

$(2x-5y)(2x+5y)$ **[A1]**

[Back to Question] *[Total Marks: 6]*

27 **(a)** $(x-4)(2x^2+7x+6)$ or $(2x^2-5x-12)(x+2)$ or $(x^2-2x-8)(2x+3)$ **[M1]**

$2x^3+7x^2+6x-8x^2-28x-24$ (allow one error in expansion) **[M1]**

$2x^3-x^2-22x-24$ **[A1]**

(b) $(4x-7y)(4x+7y)$ **[B2]**

If not B2, then B1 for $(4x-ky)(4x+ky)$ or $(kx-7y)(kx+7y)$ where $k\neq 0$ **[B1]**

(c) $(3x+2)(x-4)$ **[B2]**

If not B2, then B1 for $(3x+a)(x+b)$ where $ab=-8$ or $3b+a=-10$ **[B1]**

[Back to Question] *[Total Marks: 7]*

28 **(a)** Expands any two brackets correctly, e.g., $(2x^2-5x-12)(x+1)$ or $(x-4)(2x^2+5x+3)$ or $(x^2-3x-4)(2x+3)$ **[M1]**

Multiplies their resulting quadratic by the third bracket, allowing one error **[M1]**

$2x^3-3x^2-17x-12$ **[A1]**

(b) $(3x+a)(2x+b)$ or $(6x+c)(x+d)$ where $ab=-10$ or $cd=-10$ **[M1]**

$(3x+2)(2x-5)$ **[A1]**

(c) $x^2(x+3)-4(x+3)$ or $x(x^2-4)+3(x^2-4)$ **[M1]**

$(x^2 - 4)(x + 3)$ **[M1]**

$(x - 2)(x + 2)(x + 3)$ **[A1]**

[Back to Question] *[Total Marks: 8]*

29 **(a)** M1 for correctly expanding any pair of brackets, e.g., $2x^2 - 3x - 2$ or $2x^2 + 7x + 3$ or $x^2 + x - 6$

M1 for multiplying their quadratic expression by the third bracket, allowing one error in terms

$2x^3 + 3x^2 - 11x - 6$ **[A1]**

(b) M1 for $(3x + a)(2x + b)$ where $ab = -10$ or $3b + 2a = -11$

$(3x + 2)(2x - 5)$ **[A1]**

(c) M1 for partial factorisation extracting x: $x(3x^2 + 5x - 2)$

M1 for factorising the quadratic factor: $(3x - 1)(x + 2)$

$x(3x - 1)(x + 2)$ **[A1]**

[Back to Question] *[Total Marks: 8]*

30 **(a)** $(x^2 + x - 6)(2x - 1)$ or $(x - 2)(2x^2 + 5x - 3)$ or $(x + 3)(2x^2 - 5x + 2)$ **[M1]**

$2x^3 - x^2 + 2x^2 - x - 12x + 6$ (allow one error in expansion) **[M1]**

$2x^3 + x^2 - 13x + 6$ **[A1]**

(b) $(3x \pm a)(2x \pm b)$ where $ab = 10$ or $6x^2 - 15x + 4x - 10$ **[M1]**

$(3x + 2)(2x - 5)$ **[A1]**

(c) $2x(x^2 - 9)$ **[M1]**

$(x - 3)(x + 3)$ seen **[M1]**

$2x(x - 3)(x + 3)$ **[A1]**

[Back to Question] *[Total Marks: 8]*